

BTP Project — Cross-Dataset Generalization of ML-Based Intrusion Detection

Topic: Evaluating and improving the cross-dataset generalization of machine-learning intrusion detection systems across CIC-IDS-2017, UNSW-NB15, and TON_IoT, using metaheuristic/bio-inspired optimization for transferable feature selection.

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Document set

Submission deliverables (mid-semester)

#	Document	Purpose
05	Mid-Semester Report (18 pp)	The signable report with all required sections (title page, abstract, introduction, motivation, literature survey, problem statement, objectives, methodology, implementation, results, conclusion, references).
06	Presentation (20 slides)	Slide deck for the 10-minute presentation. Exports: .pptx (editable), .pdf (handout).

Preparation / understanding

#	Document	Purpose
07	What We Are Doing (plain language)	Team explainer: the whole project in simple words, analogies, glossary.
08	Viva Q&A Prep	55 likely questions with model answers (including per-paper questions).

Supporting analysis

#	Document	Purpose
01	Dataset Report	First-hand analysis of the three datasets, feature-schema mismatch, taxonomy mapping.
02	Literature Review (15 papers)	Focused review with detailed per-paper summaries and the gap-synthesis table.
03	Research Gap, Objectives & Scope	Formal gap statements, novelty, objectives, scope, risks.
04	Methodology & Plan	7-phase plan, evaluation protocol, fitness-function design, timeline.

Figures are in figures/. Each .md also has .docx and .pdf exports.

Supporting code

- `src/analyze_datasets.py` — computes dataset statistics from the raw CSVs.
- `src/generate_report_assets.py` — regenerates `dataset_stats.json` and the class-distribution figures.
- `src/generate_extra_figures.py` — generates the pipeline, Gantt and shared-subspace figures.
- `src/baseline_cross_dataset.py` — trains the baseline models and produces 3×3 cross-dataset metric matrices.
- `src/aggregate_baseline.py` — aggregates per-model matrices into `results/baseline_summary.md`.
- `src/baseline_figures.py` — generates the baseline-result figures.
- `src/metaheuristic_fs.py` — GA + binary PSO with a **cross-dataset fitness** (raw-feature baseline attempt).
- `src/domain_aligned_fs.py` — **DA-MFS**: domain alignment + GA/PSO feature selection (main result).
- `src/dataset_stats.json` — measured statistics (rows, class distributions, NaN/Inf counts).

Baseline result (full datasets)

Five ensemble models trained on the **entire** datasets:

Model	Same-dataset Acc	Cross-dataset Acc	Cross-dataset Bal-Acc
Random Forest	0.9666	0.3726	0.4650
XGBoost	0.9622	0.4150	0.4760
Hist. Grad. Boosting	0.9638	0.4179	0.4941
Extra Trees	0.9602	0.4027	0.4672
Soft-voting	0.9656	0.3968	0.4624

Cross-dataset balanced accuracy $\approx 0.46\text{--}0.49$ (random chance = 0.50), reproducing the generalisation gap of [1], [2].

DA-MFS optimisation result (main contribution)

Plain feature selection on raw features barely helps, because the same feature has a different distribution in each dataset. **DA-MFS** first aligns each dataset’s feature marginals (per-dataset quantile transform $\rightarrow$ normal) and then uses a Genetic Algorithm with a **cross-dataset fitness** to select a transferable subset. Full-data cross-dataset results (mean over 6 train $\rightarrow$ test pairs):

Configuration	Accuracy	Balanced acc	Macro-F1	Attack recall
Raw, all 12 features	0.3726	0.4650	0.3076	0.4789
Aligned, all 12 features	0.4776	0.5298	0.3845	0.6614
DA-MFS (aligned + 4 selected)	0.5083	0.5689	0.4604	0.6065

Improvement over raw baseline: accuracy **+13.6 pp**, balanced accuracy **+10.4 pp**, macro-F1 **+15.3 pp**, attack recall **+12.8 pp**. Selected features: duration, src_pkts, total_bytes, total_bytes_per_pkt.

Quick facts (measured from the downloaded data)

- CIC-IDS-2017: 2,830,743 flows, 77 unique features, 80.3% benign, 1,358 NaN + 4,376 Inf cells.
- UNSW-NB15: 257,673 records, 42 features, 9 attack classes + Normal.
- TON_IoT: 211,043 records, 42 features, 10 attack classes (near-balanced).
- The three datasets share **no common column names**; only ~6 raw features are semantically alignable.

How to read this for the review

1. Start with **07** (plain-language overview), then **05** (the report).
2. **08** for answering questions.
3. **01–04** for the detailed analysis behind the report.